Ch. 3.7 #4, 7 and Applied Exercise 10

4.

(a)

The training RSS for the cubic model should be lower (or at worst, equal to). The linear model is nested within the cube one (where we set both coeffecients to 0), and the lesat squares minimizes RSS over all the parameters. So the cubic fit can never do worse on the training data. It should fit some of the noise too, which will push RSS down further.

(b)

The linear model should have a lower test RSS. The cubic model has overfit the noise in the training set, so it will add variance without reeducing bias.

(c)

The cubic model has lower training RSS because it has the same nesting behavior, unlike the linear.

7.



10.

In [3]:

```
install.packages("ISLR2")
library(ISLR2)   # or library(ISLR); both contain Carseats
data(Carseats)
```

```
The downloaded binary packages are in
        /var/folders/rk/hm976cm116b00fkw44hrlkgc0000gn/T//RtmpyO4FHe/downloaded_packages
```

(a)

In [4]:

```
fit_full <- lm(Sales ~ Price + Urban + US, data = Carseats)
summary(fit_full)
```

```
Call:
lm(formula = Sales ~ Price + Urban + US, data = Carseats)

Residuals:
    Min      1Q  Median      3Q     Max
-6.9206 -1.6220 -0.0564  1.5786  7.0581

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.043469   0.651012  20.036  < 2e-16 ***
Price       -0.054459   0.005242 -10.389  < 2e-16 ***
```

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```
UrbanYes    -0.021916   0.271650  -0.081    0.936
USYes        1.200573   0.259042   4.635 4.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.472 on 396 degrees of freedom
Multiple R-squared:  0.2393,    Adjusted R-squared:  0.2335
F-statistic: 41.52 on 3 and 396 DF,  p-value: < 2.2e-16
```

(b)

In [5]:

```r
# Check how R coded the qualitative variables (reference levels are "No")
contrasts(Carseats$Urban)
contrasts(Carseats$US)

coef(fit_full)
# Price:    change in Sales (thousands) per $1 increase in Price, holding Urban and US f
# UrbanYes: urban minus rural difference in Sales, holding Price and US fixed
# USYes:    US minus non-US difference in Sales, holding Price and Urban fixed
```

A matrix: 2 × 1 of type dbl

	Yes
No	0
Yes	1

A matrix: 2 × 1 of type dbl

	Yes
No	0
Yes	1

(Intercept): 13.0434689367649 **Price:** -0.0544588491775822 **UrbanYes:** -0.0219161508141396 **USYes:** 1.20057269779412

(c)

In [6]:

```r
b <- round(coef(fit_full), 4)
cat(sprintf(
  "Sales = %.4f + (%.4f)*Price + (%.4f)*Urban[Yes] + (%.4f)*US[Yes]\n",
  b["(Intercept)"], b["Price"], b["UrbanYes"], b["USYes"]
))
# Urban[Yes] = 1 if the store is urban, 0 if rural
# US[Yes]    = 1 if the store is in the US, 0 otherwise
```

```
Sales = 13.0435 + (-0.0545)*Price + (-0.0219)*Urban[Yes] + (1.2006)*US[Yes]
```

(d)

In [7]:

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```
coef_tab <- summary(fit_full)$coefficients
coef_tab
coef_tab[, "Pr(>|t|)"] < 0.05   # TRUE = reject H0 at the 5% level
```

A matrix: 4 × 4 of type dbl

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	13.04346894	0.651012245	20.03567373	3.626602e-62
Price	-0.05445885	0.005241855	-10.38923205	1.609917e-22
UrbanYes	-0.02191615	0.271650277	-0.08067781	9.357389e-01
USYes	1.20057270	0.259041508	4.63467306	4.860245e-06

(Intercept): TRUE **Price:** TRUE **UrbanYes:** FALSE **USYes:** TRUE

(e)

In [8]:

```
fit_small <- lm(Sales ~ Price + US, data = Carseats)
summary(fit_small)
```

```
Call:
lm(formula = Sales ~ Price + US, data = Carseats)

Residuals:
    Min      1Q  Median      3Q     Max
-6.9269 -1.6286 -0.0574  1.5766  7.0515

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.03079    0.63098  20.652  < 2e-16 ***
Price       -0.05448    0.00523 -10.416  < 2e-16 ***
USYes        1.19964    0.25846   4.641 4.71e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.469 on 397 degrees of freedom
Multiple R-squared:  0.2393,    Adjusted R-squared:  0.2354
F-statistic: 62.43 on 2 and 397 DF,  p-value: < 2.2e-16
```

(f)

In [9]:

```
fit_stats <- function(m) {
  s <- summary(m)
  c(R2 = s$r.squared,
    Adj_R2 = s$adj.r.squared,
    RSE = s$sigma,
    F = unname(s$fstatistic["value"]))
}
rbind(full = fit_stats(fit_full), small = fit_stats(fit_small))

anova(fit_small, fit_full)   # F-test: does adding Urban improve the fit?
```

A matrix: 2 × 4 of type dbl

	R2	Adj_R2	RSE	F

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	R2	Adj_R2	RSE	F
full	0.2392754	0.2335123	2.472492	41.51877
small	0.2392629	0.2354305	2.469397	62.43114

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	397	2420.874	NA	NA	NA	NA
2	396	2420.835	1	0.03979039	0.00650891	0.9357389

(g)

In [10]:

```
confint(fit_small, level = 0.95)
```

A matrix: 3 × 2 of type dbl

	2.5 %	97.5 %
(Intercept)	11.79032020	14.27126531
Price	-0.06475984	-0.04419543
USYes	0.69151957	1.70776632

(h)

In [11]:

```
par(mfrow = c(2, 2))
plot(fit_small)
par(mfrow = c(1, 1))

# Outliers: studentized residuals beyond +/- 3
stud_res <- rstudent(fit_small)
max(abs(stud_res))
which(abs(stud_res) > 3)

# High leverage: compare to the average leverage (p + 1) / n
lev <- hatvalues(fit_small)
avg_lev <- mean(lev)                    # equals (p + 1) / n = 3 / 400
max(lev)
which(lev > 3 * avg_lev)                # common rule-of-thumb cutoff
plot(lev, type = "h", ylab = "Leverage"); abline(h = 3 * avg_lev, lty = 2)

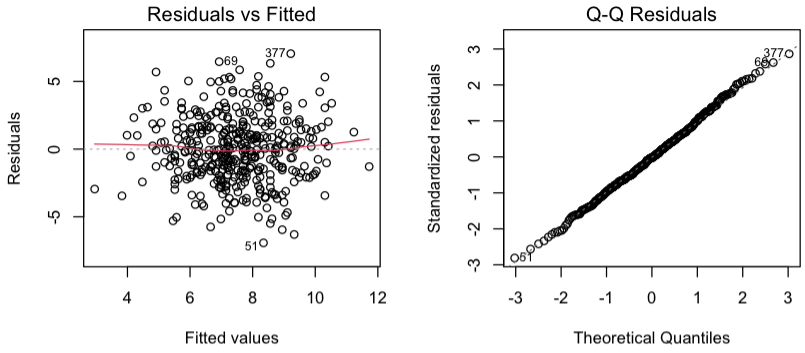
# Influence: Cook's distance (values near or above 1 are a concern)
cd <- cooks.distance(fit_small)
max(cd)
which.max(cd)
```

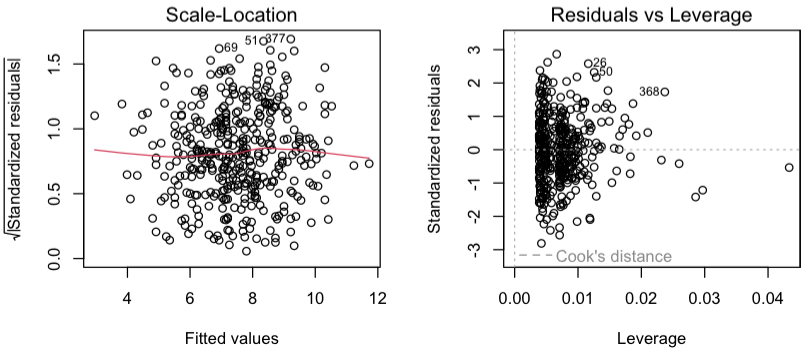
2.89152131863043

0.0433376570371785

43: 43 **126:** 126 **166:** 166 **175:** 175 **314:** 314 **368:** 368

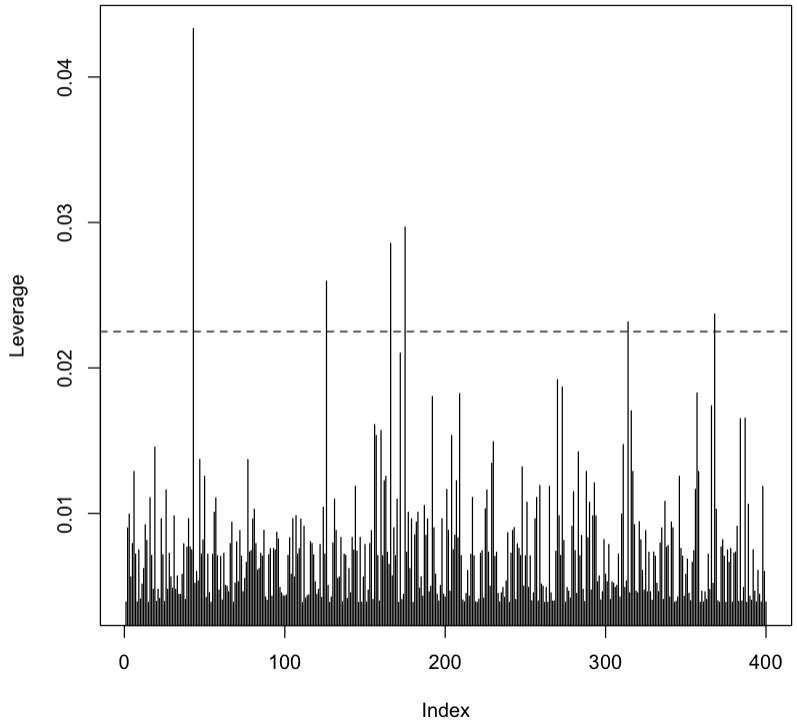
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0.0261094571634183

26: 26



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